



What is ELK Stack?

Agenda

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Need For Log Analysis

02

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What Is ELK Stack?

04

Features Of ELK Stack

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Companies Using ELK Stack

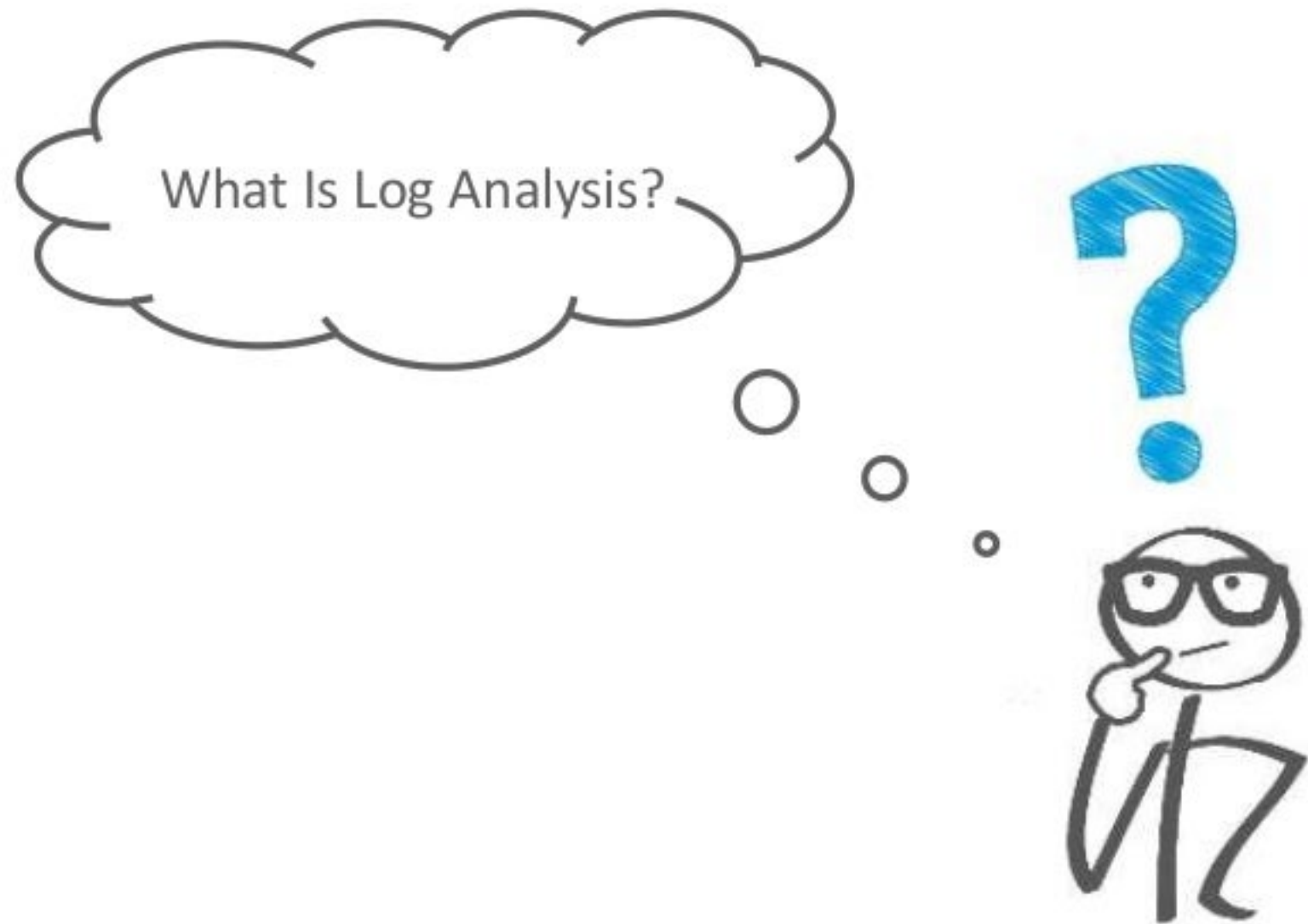




Need For Log Analysis

Let's understand why do we need Log Analysis?

What Is Log Analysis?



What Is Log Analysis?

Log Analysis Is The Process Of Analyzing The Computer/ Machine Generated Data



Need For Log Analysis

Issue Debugging



Security Analysis



Internet of Things &
Debugging



Predictive Analysis



Performance Analysis





Problems With Log Analysis

Let's understand what problems occurred with Log Analysis

Problems With Log Analysis



1

Non-consistent log format

2

Non-consistent time format

3

Decentralized logs

4

Expert knowledge requirement

Problems With Log Analysis

1 Non-consistent log format

Tomcat Logs

```
ffMay 24, 2015 3:56:26 PM org.apache.catalina.startup.HostConfig deployWAR  
INFO: Deployment of web application archive \soft\apache-tomcat-7.0.62\webapps\sample.war  
has finished in 253 ms
```

2 Non-consistent time format

Apache Access Logs

```
127.0.0.1 - - [24/May/2015:15:54:59 +0530] "GET /favicon.ico HTTP/1.1" 200 21630
```

3 Decentralized logs

IIS Logs

```
2012-05-02 17:42:15 172.24.255.255 - 172.20.255.255 80 GET /images/favicon.ico - 200  
Mozilla/4.0+(compatible;MSIE+5.5;+Windows+2000+Server)
```

4 Expert knowledge requirement

Problems With Log Analysis

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Non-consistent log format

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Non-consistent time format

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Decentralized logs

4

Expert knowledge requirement

- 142920788
- Oct 12 23:21:45
- [5/May/2015:08:09:10 +0000]
- Tue 01-01-2009 6:00
- 2015-05-30 T 05:45 UTC
- Sat Jul 23 02:16:57 2014
- 07:38, 11 December 2012 (UTC)

Problems With Log Analysis

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Non-consistent log format

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Non-consistent time format

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Decentralized Logs

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Expert knowledge requirement



Problems With Log Analysis

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Decentralized logs

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Expert Knowledge Requirement

- Everyone do not have access to the logs
- General people might not have technical expertise to understand the information
- This can slow down the analysis process



Log Management Tools

splunk>

graylog



 logentries™

 sumologic

LOGGLY

Log Management Tools

splunk>

graylog



 logentries™

 sumologic

LOGGLY



What Is ELK Stack?

Let's now understand what exactly is ELK Stack.

What Is ELK Stack?

ELK Stack is a combination of **three** open source tools which forms a **log management tool/ platform**, that helps in deep *searching, analyzing* and *visualizing* the log generated from different machines



elasticsearch



logstash



kibana

What Is ELK Stack: ElasticSearch



elasticsearch



Features

- ✓ search engine/ search server
- ✓ NoSQL database i.e. can't use SQL for queries.
- ✓ Based on Apache Lucene and provides RESTful API
- ✓ Provides horizontal scalability, reliability and multitenant capability for real time search
- ✓ Uses indexes to search which makes it faster

ORACLE[®]
NO SQL
DATABASE



riak



APACHE
HBASE



mongoDB.

What Is ELK Stack: ElasticSearch



elasticsearch



Companies Using ElasticSearch

guardian



StumbleUpon



WIKIPEDIA
The Free Encyclopedia

GitHub



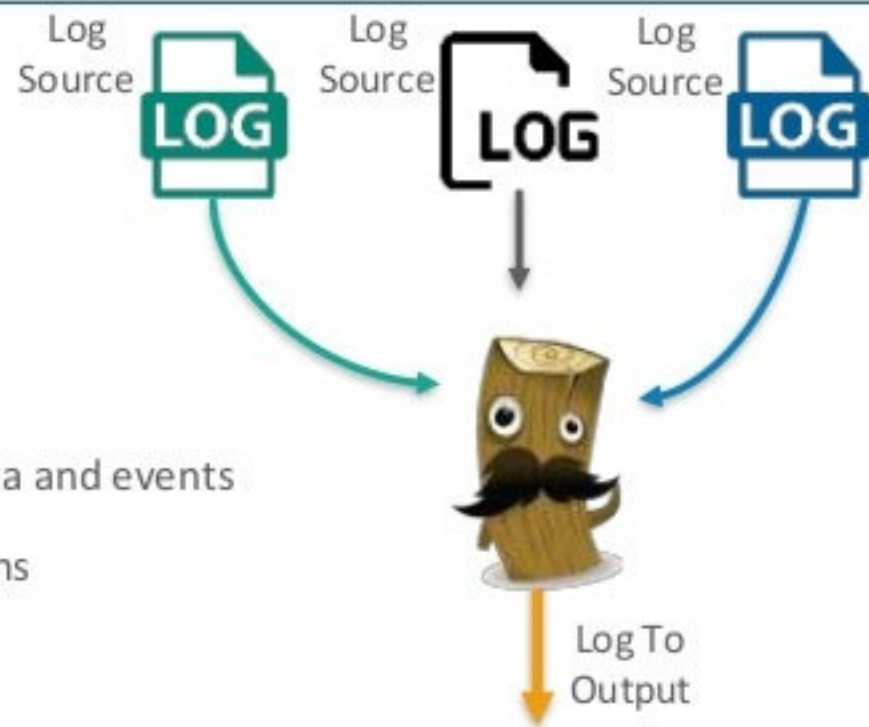
SOUNDCLOUD

What Is ELK Stack: LogStash

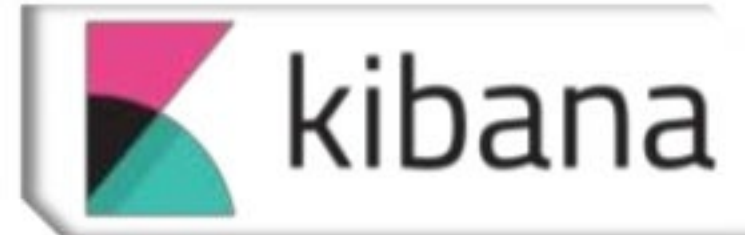


Features

- ✓ Data pipeline tool
- ✓ Centralizes the data processing
- ✓ Collects, parses and analyzes large variety of structured/ unstructured data and events
- ✓ Provides plugins to connect to various types of input sources and platforms



What Is ELK Stack: Kibana



Features

- ✓ Visualization tool
- ✓ provides real-time analysis, summarization, charting, and debugging capabilities.
- ✓ Provides instinctive and user friendly interface
- ✓ Allows sharing of snapshots of the logs searched through.
- ✓ Permits saving the dashboard and managing multiple dashboards

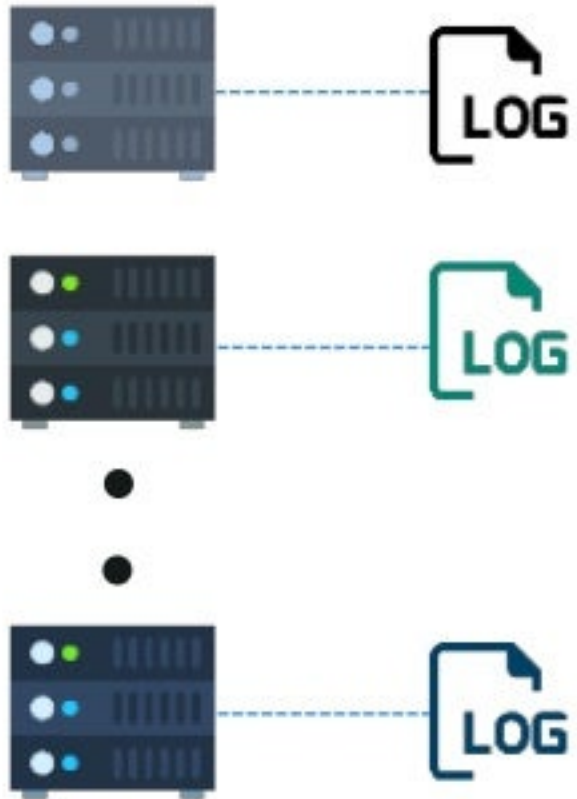




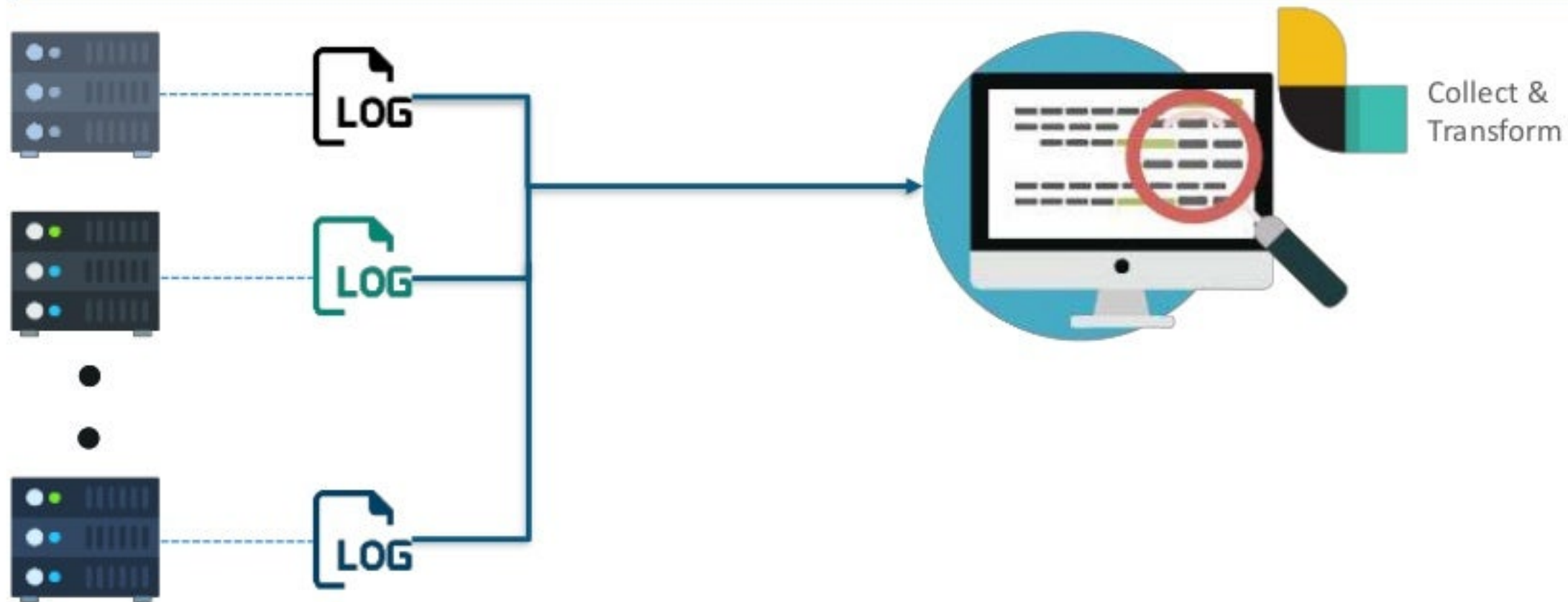
How ELK Stack Works?

Let's now find out how ELK works together.

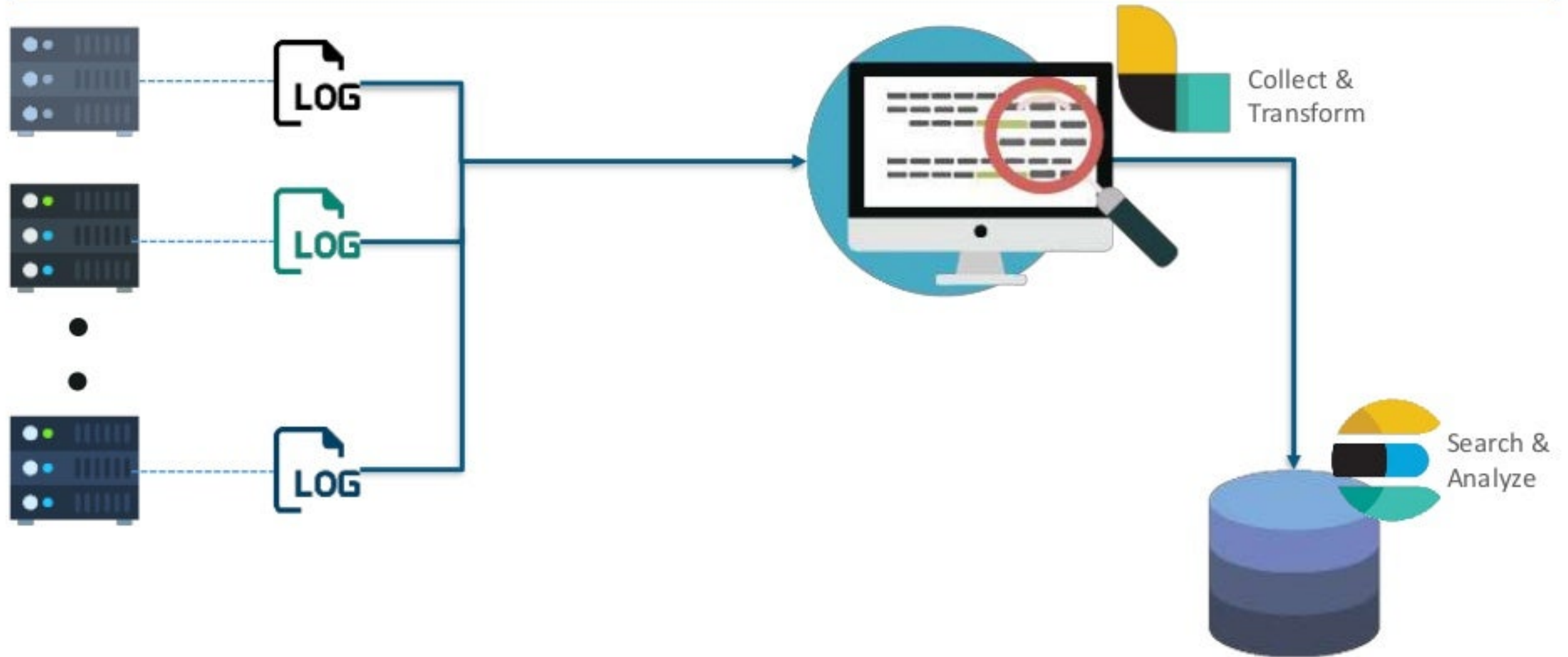
How ELK Stack Works?



How ELK Stack Works?



How ELK Stack Works?



How ELK Stack Works?





Companies Using ELK Stack

Companies Using ELK Stack

